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Review

Prostate Magnetic Resonance Imaging Using the Prostate Imaging for Recurrence Reporting (PI-RR) Scoring System to Detect Recurrent Prostate Cancer: A Systematic Review and Meta-analysis

Felipe A. Mourato^{a,*}, Luiza G. Schmitt^b, Miriana Mariussi^c, Giovanni Torri^d, Stephan Altmayer^e, Francesco Giganti^{f,g}, Jorge Abreu-Gomez^h, Nathan Perlisⁱ, Alejandro Berlin^j, Sangeet Ghai^h, Masoom A. Haider^h, Adriano B. Dias^h

^a Unidade de Diagnóstico por Imagem, Empresa Brasileira de Serviços Hospitalares, Hospital das Clínicas da Universidade Federal de Pernambuco, Recife, Brazil;

^b Department of Radiation Oncology, UT Southwestern, Dallas, TX, USA; ^c Department of Diagnostic Radiology, Hospital Universitario Austral, Buenos Aires, Argentina;

^d Department of Radiology and Diagnostic Imaging, Hospital Universitário de Santa Maria, Universidade Federal de Santa Maria, Santa Maria, Brazil; ^e Department of Radiology, Stanford University School of Medicine, Stanford, CA, USA; ^f Department of Radiology, University College London Hospital NHS Foundation Trust, London, UK; ^g Division of Surgery and Interventional Science, UCL, London, UK; ^h University Medical Imaging Toronto; Joint Department of Medical Imaging; University Health Network–Sinai Health System–Women's College Hospital, University of Toronto, Toronto, ON, Canada; ⁱ Division of Urology, Department of Surgery, Princess Margaret Cancer Centre, University Health Network, Toronto, Canada; ^j Department of Radiation Oncology, Princess Margaret Cancer Center, University Health Network and University of Toronto, Toronto, Canada

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Abstract

Background and objective: Prostate Imaging for Recurrence Reporting (PI-RR) was introduced in 2021 to standardize the interpretation and reporting of multiparametric magnetic resonance imaging (MRI) for prostate cancer following whole-gland treatment. The system scores image on a scale from 1 to 5 and has shown promising results in single-center studies. The aim of our systematic review and meta-analysis was to assess the diagnostic performance of the PI-RR system in predicting the likelihood of local recurrence after whole-gland treatment.

Methods: The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for diagnostic test accuracy were followed. Relevant databases were searched up to December 2023. Primary studies met the eligibility criteria if they reported MRI diagnostic performance in prostate cancer recurrence using PI-RR. Diagnostic performance for MRI was assessed using two different cutoff points (≥ 3 or ≥ 4 for positivity according to the PI-RR system). A meta-analysis with a random-effects model was used to estimate pooled sensitivity and specificity values.

Key findings and limitations: Sixteen articles were identified for full-text reading, of which six were considered eligible, involving a total of 467 patients. Using a cutoff of PI-RR ≥ 3 (4 studies) for recurrent disease, the sensitivity was 77.8% (95% confidence interval [CI] 69.9–84.1%) and the specificity was 80.2% (95% CI 58.2–92.2%). Using a cutoff of PI-RR ≥ 4 (4 studies), the sensitivity was 61.9% (95% CI 35.6–82.7%) and the

* Corresponding author. Setor de Medicina Nuclear, Hospital das Clínicas da Universidade Federal de Pernambuco, Avenida Prof. Moraes Rego 1235, Cidade Universitária, Recife, PE CEP 50670-901, Brazil. Tel. +55 81 21263633.

E-mail address: mourato.fa.ciencia@gmail.com (F.A. Mourato).

specificity was 86.6% (95% CI 75.1–93.3%). Overall, the inter-rater agreement varied from fair to excellent.

Conclusions and clinical implications: PI-RR is accurate in detecting local recurrence after whole-gland treatment for prostate cancer and shows fair-to-good to excellent inter-reader agreement. Overall, a PI-RR cutoff of ≥ 3 showed high sensitivity and specificity.

Patient summary: We reviewed studies that reported on how good MRI scans using a scoring system called PI-RR were in detecting recurrence of prostate cancer. We found that this system shows good performance, with fair to excellent agreement between different radiologists.

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1. Introduction

Prostate cancer is a public health problem as it is one of the most frequently diagnosed malignancies and a leading cause of mortality worldwide [1]. The main therapeutic options for localized disease include primary radiotherapy (RT) and radical prostatectomy (RP). However, approximately 20–50% of these patients experience biochemical recurrence (BCR) by 10 years [2–4]. In these cases, accurate diagnosis and localization of disease recurrence is paramount for effective patient management and guiding appropriate salvage treatment.

The imaging modalities available offer complementary roles. According to current guidelines, prostate-specific membrane antigen (PSMA) positron emission tomography/computed tomography (PET/CT) is the most suitable imaging modality for detection of distant metastasis [5]. Conversely, for detection of local recurrences that could be amenable to curative-intent treatments, it has been demonstrated that multiparametric magnetic resonance imaging (mpMRI) of the prostate is accurate in both post-RT and post-RP settings [6]. The European Association of Urology guidelines recommend the use of mpMRI and PET/CT for cases of BCR after RT. In addition, guidelines from the American Society for Radiation Oncology, American Urological Association, and the National Comprehensive Cancer Network recommend mpMRI in patients experiencing BCR after RP to assess for local recurrence [7–9].

The Prostate Imaging for Recurrence Reporting (PI-RR) scoring system was introduced in 2021 to standardize mpMRI interpretation and reporting in prostate cancer cases following whole-gland treatment (RT or RP) [10]. Individual studies that evaluated the performance of mpMRI using PI-RR have shown encouraging results [11–16]. The PI-RR system uses a 5-point scale to classify and document local recurrence in patients with prostate cancer patient [10]. A lesion with a score of 1 or 2 points is defined as a very low or low likelihood of recurrence; a score of 3 points is defined as uncertain; and a score of 4 or 5 points is defined as a high or very high likelihood of recurrence.

To the best of our knowledge, there have been no meta-analyses on the performance of the PI-RR system in identifying prostate cancer recurrence. The aim of our systematic review and meta-analysis was to assess the diagnostic performance of the PI-RR system in detecting the presence of

local recurrence of prostate cancer after whole-gland treatment.

2. Methods

2.1. Search strategy

Our study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines on diagnostic test accuracy [17]. The literature was searched up to December 2023 in the PubMed/MEDLINE, EMBASE, and Cochrane databases. The search strategy for each database is provided in Supplementary Table 1. The review was registered in the International Prospective Register of Systematic Reviews (PROSPERO; CRD42024506028).

2.2. Inclusion and exclusion criteria

We included studies that evaluated the diagnostic performance of the PI-RR system for detection of prostate cancer recurrence after whole-gland therapy. We excluded case reports, case series, letters to the editor, and review articles. There were no language restrictions. The population, index test, and target condition approach was used to define study eligibility according to the Cochrane handbook on systematic reviews of diagnostic test accuracy [18]. The terms used in the database searches are included in Supplementary Table 1.

2.3. Assessment of study quality

The quality of all eligible studies was assessed by two reviewers independently using the Quality Assessment of Diagnostic Studies (QUADAS)-2 tool [19]. In cases of any discrepancy between the two reviewers, a third reviewer was involved to resolve the issue via consensus.

2.4. Data extraction

Two reviewers conducted the study selection independently and extracted relevant data from the selected studies into a standardized form, including study characteristics, demographics, and diagnostic performance of MRI (including true positive, false positive, true negative, and false negative rates). Any disagreement between the two reviewers was resolved via consensus with the assistance of a third reviewer.

2.5. Statistical analysis

Diagnostic performance for MRI was assessed using two different cutoff points ($PI-RR \geq 3$ or ≥ 4) to define imaging-predicted disease recurrence. In studies with multiple readers, the consensus result was used for meta-analysis. If consensus values were not reported, the average results among the readers were calculated for true positive, false positive, true negative, and false negative rates for use in the meta-analysis. When available, the diagnostic performance per

MRI sequence (T2-weighted imaging [T2WI], dynamic contrast enhancement [DCE], and diffusion-weighted imaging [DWI]) was also included.

Pooled sensitivity and specificity values with 95% confidence intervals (CIs) were calculated (random-effect analysis). Analysis was performed to investigate potential reasons for heterogeneity. Inter-rater agreement was recorded and Cohen's κ was interpreted according to previous reports [20]. All analyses were performed using Rstudio 2023.09.1 with R v4.3.2 and the *mada* and *meta* packages.

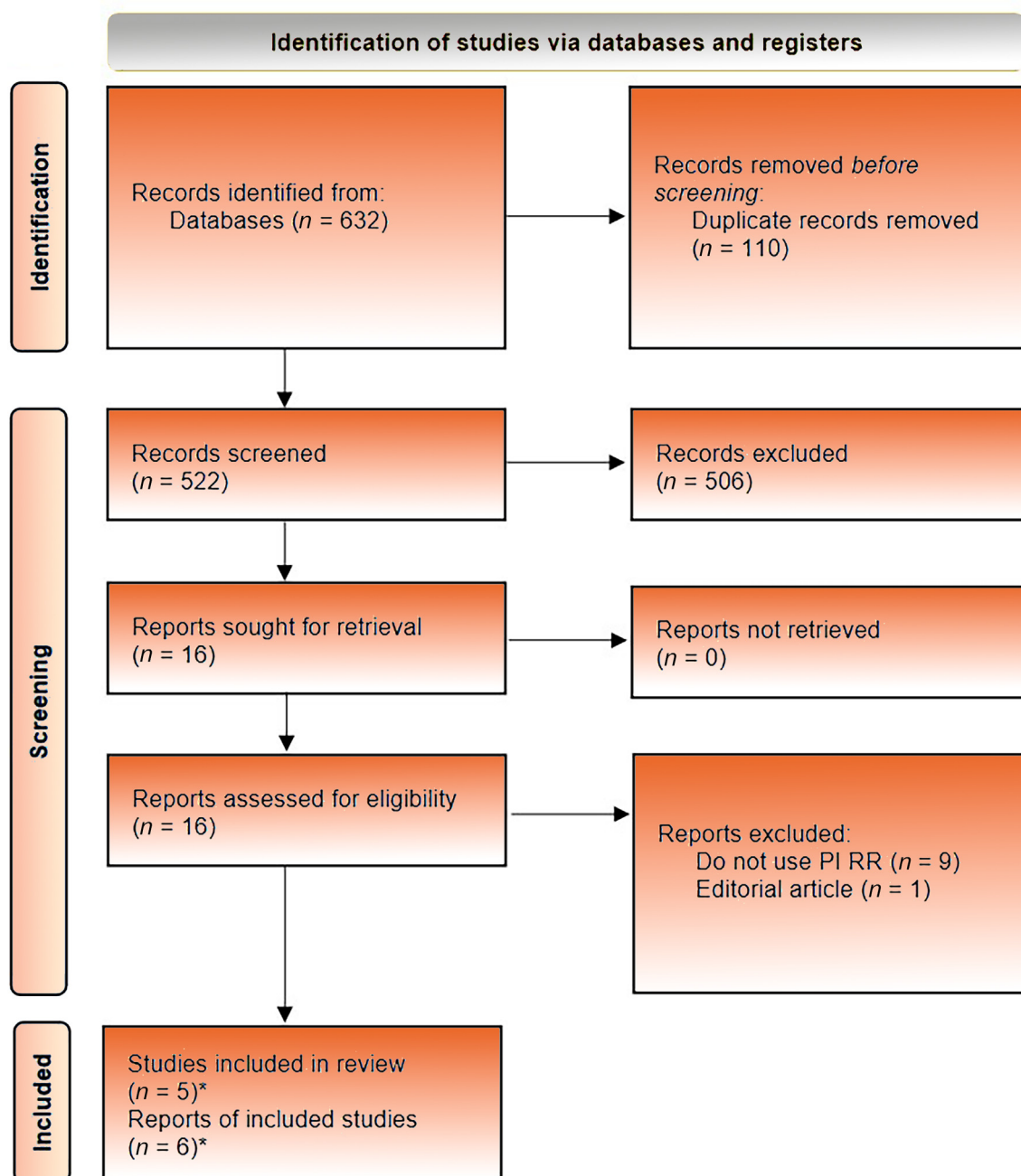


Fig. 1 – Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart for study selection. * Park et al [16] was only included in the qualitative analysis.

Table 1 – Main characteristics of the studies included in the review

Study and location	Design	TSA (d)	Pts	Patient age (yr)	Time since iTx	Reference standard	Prostate-specific antigen (ng/ml)	
							Before iTx (ng/ml)	MRI or BCR
Bergaglio 2023 [11] Italy	SCR	35	76 RT: 8 RP: 68	72.4 ^a 73.4 (76.1–69.1) ^b	5.1 (8.51.5) ^b	Correlative imaging, PSA after Tx/FU, new lesions on subsequent imaging	NA	1.01 ^{a,e} 0.27 (0.21–0.76) ^{b,e}
Ciccarese 2022 [12] Italy	SCR	23	19 All RP	69.2 ± 6.6 ^c	NA	Lesion Bx	NA	0.79 (0.0–5.85) ^f
Franco 2023 [13] Spain, Italy, and Portugal	SCR	271	120 RT: 12 RP: 108	72 (62–82) ^b	2.5 (0.2–17.2) ^d	Lesion Bx, PSA after Tx/FU, lesion size at FU	7.34 (4.3–41.2) ^b	0.32 (0.18–5.38) ^d
Pecoraro 2022 [14] Italy, USA, Belgium, and Netherlands	SCR	179	100 RT: 48 RP: 52	RT: 76 (70–82) ^b RP: 70 (66–74) ^b	RT: 37 mo (12–78) ^b RP: 43 mo (20–82) ^b	Lesion Bx, PSA after Tx/FU, lesion size at FU	RT: 7.34 (0.02–46) ^d RP: 7.32 (1.1–28.8) ^d	RT: 1.9 (0.015–10.77) ^d RP: 0.28 (0.01–1.82) ^d
Kim 2022 [15] South Korea	SCR	46	468 All RP ^g	66.2 ± 6.9 ^c	1042.7 ± 820.5 ^c	PSA levels after Tx/FU, lesion size at FU	26.3 ± 108.2 ^c	1.3 ± 4.6 ^c
Park 2022 [16] South Korea	SCR	176	272 All RP ^a	66.6 ± 7.4 ^c	138 (102–365) ^d	HPx for surgical margin	NA	0.06 ^a

BCR = biochemical recurrence; Bx = biopsy; FU = follow-up; FU = follow-up imaging; HPx = histopathology; IQR = interquartile range; iTx = initial treatment; MRI = magnetic resonance imaging; NA = not available; PSA = prostate-specific antigen; RT = radiation therapy; RP = radical prostatectomy; SCR = single-center retrospective study; TSA = time from submission to acceptance; Tx = treatment.

Only the dynamic contrast enhancement sequence was analyzed.

^a Mean.
^b Median (interquartile range).
^c Mean ± standard deviation.
^d Median (range).
^e Results reported as ng/ml.
^f Mean (range).
^g Only 153 patients were analyzed according to the PI RR system.

Table 2 – Characteristics of the MRI protocol and reader experience in the studies

Study	Number of readers (PMRI experience)	Inter-reader agreement	Type of analysis	MRI field strength (scanner)	MRI acquisition protocol	PI-RR adherence
Bergaglio 2023 [11]	3 (3, 6, and 10 yr)	Gwet's κ = 0.74 for individual scores (good agreement).	RDP reported separately	1.5T (Magnetom AERA, Siemens)	T2WI: axial, coronal, and sagittal; DWI and DCE obtained with the same ST and plane to obtain a match DWI: SSEP sequence with a high b-value (1400 s/mm ²) and another sequence with 0, 750, 1000 s/mm ² ; the latter was used to obtain the ADC map DCE: Gd contrast agent (0.2 ml/kg) at 3 ml/s followed by 15 ml of saline solution; temporal resolution 9 s	T2W: Yes DWI: Yes DCE: Yes
Ciccarese 2022 [12]	2 (5 and 10 yr)	Reporting score agreement κ = 0.884 (almost perfect agreement)	Consensus between readers	1.5T (SignaHDxt; GE Healthcare)	T2WI: FRFSE sequences in the sagittal, axial and coronal planes, covering the prostate lodge DWI: SSEP sequence with a high b-value (2000 s/mm ²) and another with 50 and 1000 s/mm ² ; ADC map calculated at 1000 s/mm ² DCE: 3D T1-weighted ToFSGR axial sequence during i.v. injection of a Gd contrast agent at 3 ml/s followed by 15 ml of saline solution; temporal resolution 10 s; acquisitions before contrast injection were analyzed to detect foci of hemorrhage	T2W: Yes DWI: Yes DCE: Yes
Franco 2023 [13]	5 (3, 4, 5, 9, and 20 yr)	Cohen's κ 0.52–0.77 (moderate to substantial agreement)	RDP reported separately	3.0 T (Ingenia, Philips)	T2WI: sagittal, axial, coronal planes; TR/TE 2636–3542/90–100 ms; FOV 200 × 200 mm; ST 3 mm (0.3 mm gap) DWI: TR/TE 5991/90 ms; FOV 180 × 323 mm; ST 3 mm (no gap); b-values: 0, 100, 1000, 1500 s/mm ² DCE: TR/TE 4.5/2.3 ms; FOV 250 × 300 mm; ST 4 mm (–2 mm gap)	T2W: Yes DWI: Yes DCE: Yes ^a
				1.5 T (Achieva, Philips)	T2WI: sagittal, axial, and coronal; TR/TE 3505–3853/90–105 ms; FOV 240 × 240 mm; ST 3 mm (0.3 mm gap) DWI: TR/TE 5991/90 ms; FOV 180 × 323 mm; ST 3 mm (0.3 mm gap); b-value: 0, 100, 1000, 1500 s/mm ² DCE: TR/TE 5/2 ms; FOV 270 × 270 mm; ST 4 mm (–2 mm gap)	
Pecoraro 2022 [14]	4 (10, 11, 14, and 20 yr)	Cohen's κ 0.33–0.62 (fair to moderate agreement) ICC (95% CI): RT: 0.87 (0.81–0.93) RP: 0.87 (0.80–0.92)	RDP reported separately	3.0 T (Discovery MR750, GE Healthcare) 1.5 T (Achieva, Philips)	T2WI: RP: axial, sagittal, and coronal; RT: axial and coronal; TR/TE 5000/100 ms; ST 3 mm (no gap); FOV 120 × 200 mm DWI: TR/TE ≥3000/90 ms; ST 3 mm (no gap); FOV 160 × 220 mm; b-values 100, 800–1000, 2000 s/mm ² ; ADC map calculated at 2000 s/mm ² DCE: TR/TE <100/<5 ms; ST 3 mm (no gap); temporal resolution 5 s	T2W: Yes DWI: Yes DCE: Yes

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Table 2 (continued)

Study	Number of readers (PMRI experience)	Inter-reader agreement	Type of analysis	MRI field strength (scanner)	MRI acquisition protocol	PI-RR adherence
Kim 2022 [15]	2 (3 and 20 yr)	$\kappa = 0.80, p < 0.05$ (substantial agreement)	Consensus between readers	3.0 T (Achieva or Ingenia, Philips) or 1.5 T (Amira, Siemens)	T2WI: axial, sagittal and coronal; TR/TE 2500–3000/70–90 ms; ST 3 mm (gap 1 mm); FOV 160 × 160 mm; matrix 320 × 320; number of excitations 1 DWI: b-values 0, 100, 1000, 1500 s/mm ² DCE: axial 3D sequence in 137/468 patients, single phase in 331/468 patients for 180–210 s	T2W: Yes DWI: Yes DCE: Partially ^b
Park 2022 [16]	1 (5 yr)	–	NA	3.0 T (Ingenia, Philips Healthcare; Architect, GE Healthcare; Skyra, Siemens Healthcare)	T2WI: axial, sagittal, and coronal; TR/TE 2400–2700/100 ms; ST 3 mm; FOV 180 × 180 mm; matrix 512 × 512 DWI: SSEP; TR/TE 3800–4000/82 ms; ST 3 mm; FOV 180 × 180 mm; matrix 176 × 176 mm; b-values 0, 50, 500, 1000 s/mm ² and a calculated image set reconstructed for 1500 s/mm ² DCE: TR/TE 5.8–6.2/1.5 ms; FOV 230 × 230 mm; acquisition time, 0, 30, 60, 90, and 120 s after contrast injection	T2W: Yes DWI: Yes DCE: Yes ^c

3D = three-dimensional; MRI = magnetic resonance imaging; PMRI = prostate MRI; T2WI = T2-weighted imaging; DWI = diffusion-weighted imaging; DCE = dynamic contrast enhancement; TR/TE = repetition time/time to echo; RT = radiation therapy; RP = radical prostatectomy; ICC = intraclass correlation coefficient; RDP = reader diagnostic performance; NA = not available; PI-RR = Prostate Imaging for Recurrence Reporting; ADC = apparent diffusion coefficient; ToFSGR = time-of-flight spoiled gradient-recalled; FOV = field of view; SSEP = single-shot echo planar; FRFSE = fast relaxation fast spin echo; ST = slice thickness i.v. = intravenous.

^a DCE reporting adhered to PI-RR, but the temporal resolution was not specified.

^b DCE was obtained in 137/468 patients. Single-phase contrast enhancement was used in 331/468 patients.

^c DCE reporting adhered to PI-RR, but the temporal resolution was limited.

3. Results

3.1. Study selection

The initial search yielded 632 articles (Fig. 1). After screening, the full text of 16 articles was assessed, of which six studies involving a total of 467 patients were considered eligible [11–16]. The characteristics of these studies are listed in Table 1 and Table 2. One study only reported

results for the DCE sequence [16] and was thus included in the systematic review but not the quantitative analysis.

3.2. QUADAS-2 assessment

QUADAS-2 results are presented in Figure 2. For all studies, patient selection and the index test were considered at low risk of bias. However, the reference standard was regarded as having some concerns of bias in two studies [11,15] as

	Risk of bias				Applicability concerns		
	Patient selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Bergaglio 2023	Green	Green	Yellow	Yellow	Green	Green	Yellow
Ciccaresse 2022	Green	Green	Green	Yellow	Green	Green	Green
Franco 2022	Green	Green	Green	Green	Green	Green	Green
Kim 2022	Green	Green	Yellow	Green	Yellow	Green	Yellow
Pecoraro 2022	Green	Green	Green	Green	Green	Green	Green

Fig. 2 – Quality Assessment of Diagnostic Studies (QUADAS)-2 results. Green = low risk of bias; yellow = some concerns.

Table 3 – Overall diagnostic performance

	All studies	PI-RR cutoff ≥ 3	PI-RR cutoff ≥ 4	
			All studies	Excluding Kim [15]
Sensitivity, % (95% CI)	70.2 (55.5–81.6)	77.8 (69.9–84.1)	61.9 (35.6–82.7)	74.5 (64.7–82.4)
Specificity, % (95% CI)	84.2 (75.1–90.4)	80.2 (58.2–92.2)	86.6 (75.1–93.3)	84.0 (62.7–94.3)
DOR (95% CI)	12.1 (6.3–23.5)	16.9 (8.6–33.1)	9.6 (3.2–28.8)	17.3 (7.4–40.6)

CI = confidence interval; DOR = diagnostic odds ratio; PI-RR = Prostate Imaging for Recurrence Reporting.

they did not include biopsy results. Overall, two studies had some concerns of bias [11,15] and three had a low risk of bias [12–14].

3.3. Diagnostic accuracy of PI-RR

The studies described analyses on a per-patient basis. Three studies reported the diagnostic performance of PI-RR for both cutoff values [12–14]. Only one study reported the accuracy for PI-RR ≥ 3 [11], while another study only reported results for PI-RR ≥ 4 [15].

Using a cutoff of PI-RR ≥ 3 (4 studies), the pooled sensitivity was 70.2% (95% CI 55.5–81.6%) and the pooled specificity was 84.2% (95% CI 75.1–90.4%). For a cutoff of PI-RR ≥ 4 (4 studies), the pooled sensitivity was 61.9% (95% CI 35.6–82.7%) and the pooled specificity was 86.6% (95% CI 75.1–93.3%; Table 3).

3.4. Performance of PI-RR by sequence

Only Pecoraro et al [14] reported the diagnostic performance per MRI sequence. Overall, DCE sequences performed better than T2WI and DWI for detection of recurrence after both RT and RP. Park et al [16] reported high diagnostic performance for DCE alone, with sensitivity of 84.2% and specificity of 82.3% using a cutoff of DCE ≥ 3 .

3.5. Heterogeneity analysis

In the pooled analysis for PI-RR ≥ 3 , there was no significant heterogeneity for sensitivity and there was moderate heterogeneity for specificity (Fig. 3). The latter finding seemed to be driven by one of the studies [12] and could be partly related to its small sample size (19 patients).

For the analysis using PI-RR ≥ 4 , there was high heterogeneity for sensitivity but not for specificity (Fig. 3). The

heterogeneity for sensitivity was driven by one study [15] that reported substantially lower sensitivity than the other studies, which is likely to be related to population selection, as most patients (53.6%) had a prostate-specific antigen (PSA) level lower than 0.4 ng/ml. However, the other studies only reported PSA as the median and corresponding range or interquartile range, so it was not possible to further assess the contribution of this or other factors to the heterogeneity observed. In addition, imaging for most of the patients included in the study by Kim et al [15] did not meet the PI-RR standard for the DCE sequence (Table 2), which might also have contributed to heterogeneity.

Additional subgroup analysis was not feasible given the limited number of studies.

3.6. Inter-rater agreement

All studies with more than one reader reported on inter-rater agreement [11–15], which ranged from fair-to-good to excellent (Table 2). Four studies [12–15] reported Cohen's κ coefficients, which ranged from 0.33 [14] to 0.88 [12] (fair to excellent). One study reported a Gwet's κ value of 0.74 for three readers (good agreement) [11].

4. Discussion

Our systematic review and meta-analysis comprising six studies and 467 patients demonstrated that the PI-RR scoring system has high sensitivity and specificity for detection of local recurrence after RT or RP for prostate cancer. Our findings indicate that PI-RR retains the accuracy of MRI in detecting local tumor recurrence while providing a structured assessment score. Notably, a cutoff point of 3 (on a 5-point scale) yields high sensitivity and specificity for diagnosing recurrent disease.

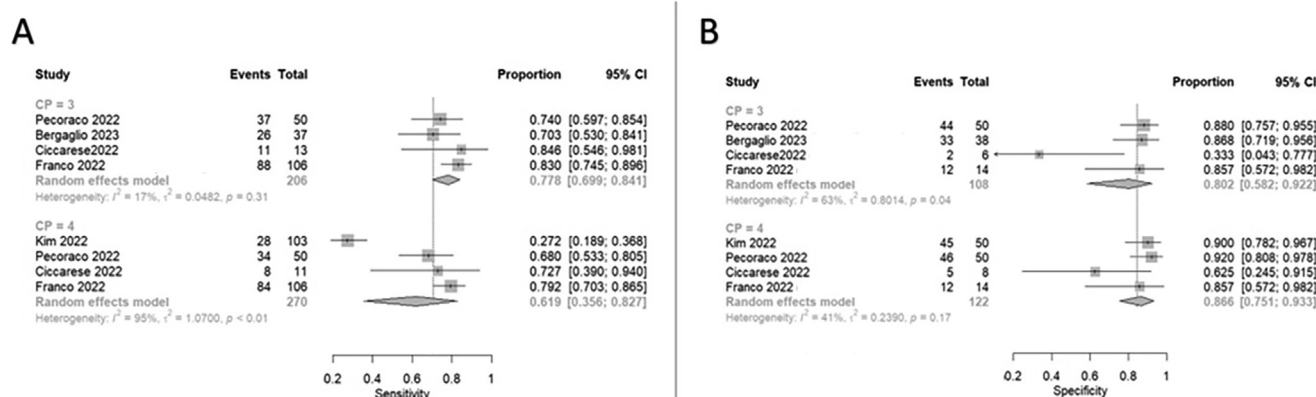


Fig. 3 – Forest plots of (A) sensitivity and (B) specificity of the Prostate Imaging for Recurrence Reporting (PI-RR) scoring system. CP = cutoff point.

A potential challenge with the scheme arises from classification of PI-RR 3 lesions as uncertain, leading to diagnostic dilemmas and impacting clinical decision-making. We assessed the diagnostic performance using two different cutoffs ($\text{PI-RR} \geq 3$ and ≥ 4). We noted greater sensitivity at the threshold of $\text{PI-RR} \geq 3$, with comparable specificity. This implies that classifying lesions as positive at $\text{PI-RR} \geq 3$ may be the most efficient strategy.

PI-RR recommends similar patient preparation, MRI equipment, and imaging protocols to those described in Prostate Imaging-Reporting and Data System v2.1 [21]. Analysis of T2WI, DCE, and DWI sequences is necessary to assess the likelihood of local recurrence. However, only one study reported the diagnostic accuracy per MRI sequence [14] and showed that DCE performed better in both RT and RP patients. These findings were expected, as it has been reported that DCE is highly sensitive in evaluating local recurrence after RP, identifying small lesions in the prostate bed even at low PSA levels [22,23]. In line with this, Park et al [16] reported excellent DCE performance in relation to the presence and location of positive surgical margins. Furthermore, the authors also observed an association between focal nodular enhancement on DCE sequences at postoperative MRI among patients without BCR at the time, and demonstrated that this finding was correlated with a shorter time to subsequent BCR.

A standardized MRI protocol and reporting system can enhance the quality of imaging examinations, improve the consistency of clinical management, and facilitate multicenter research [24]. A standardized protocol can also improve MRI reports and satisfaction among referring providers [25]. Systematization can lead to better concordance among different readers, which is crucial for utility in everyday practice [26]. The studies included in our meta-analysis showed a high range of interobserver agreement, varying from fair to excellent (Table 2). It is important to emphasize that these studies predominantly involved experienced genitourinary radiologists, so the reproducibility of results in centers with less expertise is still unknown.

Previous systematic reviews have shown that MRI is a valuable tool for detecting local recurrence after RP or RT, with high diagnostic accuracy and the potential to differentiate tissue types (such as scar/fibrotic tissue and tumor recurrence), even at low PSA levels [6,27–30]. However, these reviews also highlight significant differences regarding the MRI sequences used among studies. For instance, the systematic review by Barchetti and Panebianco [29] revealed that most studies did not include all three sequences (T2WI, DCE, and DWI) endorsed by the PI-RR system. Consequently, studies conducted before the PI-RR system exhibit considerable heterogeneity in imaging acquisition, which complicates extrapolation of diagnostic study findings to contemporary clinical practice. Our systematic review reaffirms the high diagnostic performance of mpMRI for detection of recurrent prostate cancer when adhering to the PI-RR system.

It is noteworthy that the emergence of PSMA tracers such as ^{68}Ga -PSMA-11 and ^{18}F -piflufolstat (^{18}F -DCFPyL) marks a transformative shift in the imaging landscape for BCR of prostate cancer. While PSMA PET/CT is valuable for detection of extrapelvic nodal and distant metastatic dis-

ease [31,32], its limitations for detection of local recurrence, particularly when not combined with mpMRI, need careful consideration. Urinary excretion of PSMA may obscure local recurrence in the prostate or prostate bed, impacting detection rates [33–35]. Freitag et al [36] reported that only half of the local recurrences seen on MRI after RP were detectable on PSMA PET. They also observed that the proximity of the recurrent lesion to the bladder was significantly associated with false-negative PSMA PET results, while the size of the recurrent lesion was not associated with false-negative PSMA PET results. This challenge is addressed by incorporating fusion PET/MRI, either through a hybrid system or retrospective fusion of separate mpMRI and PET/CT examinations. In 2022, Panebianco and Turkbey [37] proposed a risk-adapted pathway with integrated imaging for detection and localization of recurrence. According to the risk of disease progression, patients at low risk start with MRI, while those at intermediate or high risk should undergo PSMA PET/CT as a priority for effective management of recurrent prostate cancer. Our results support the use of mpMRI as a crucial tool for assessing recurrence after whole-gland treatment. Nonetheless, a comprehensive head-to-head comparison between MRI and PSMA PET is essential to determine the optimal imaging tool across diverse clinical scenarios.

Our study has some limitations. First, only six studies fulfilled the eligibility criteria, a limitation that could be attributed to the recent introduction of PI-RR [10]. Despite active research and publications on the use of MRI after whole-gland treatment, a structured assessment and reporting system was only established and made available in 2021. Second, according to PI-RR, DWI and DCE sequences are recognized as higher-yield modalities following RT, while DCE is considered the best sequence after RP. These distinctions may result in varying diagnostic accuracy of the PI-RR system between patients who have undergone RP and RT. Unfortunately, due to insufficient data, we could not perform subgroup calculations to pool performances based on treatment modality. Third, our primary analysis showed significant heterogeneity, mostly related to disparate performance in a single study [15], as demonstrated in the secondary analysis. This heterogeneity could be attributed to two factors: (1) the threshold effect, as this study exhibited one of the highest specificity and lowest sensitivity values; and (2) in most patients, single phase contrast enhancement was used instead of DCE, as recommended for PI-RR. More granular analysis of heterogeneity is limited by the number of studies available.

5. Conclusions

In conclusion, we found that PI-RR is accurate in detecting local recurrence after whole-gland treatment for prostate cancer. In particular, a cutoff point of 3 (on a 5-point scale) demonstrated high sensitivity and specificity for diagnosing recurrent disease. The inter-rater agreement among readers ranged from fair to excellent. Our meta-analysis results support the use of this standardized method for assessment and reporting of mpMRI in the post-RT or post-RP prostate

cancer setting and highlight a need for additional research with larger sample sizes, a multicenter approach, a prospective design, and extended follow-up.

Author contributions: Felipe A. Mourato had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Dias, Torri.

Acquisition of data: Mourato, Schmitt, Mariussi.

Analysis and interpretation of data: Mourato, Altmayer, Torri, Dias.

Drafting of the manuscript: Mourato, Dias.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.euo.2024.05.007>.

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